武汉理工大学学业成绩表（评优评先用）

院系：计算机智能学院　　专业：计算机(余)　　班级：计算机1804　　学号：0121814660102　　姓名：李程远

课程名称	性质	学分	成绩
2018-2019-1学期			
无机化学B	个性课程	3.0	95.7
军事训练	实践课	1.5	88
基础体育	体育必修	1.0	81
军事理论	通识必修	1.0	89.6
中国近现代史纲要	通识必修	2.5	91
大学英语D1	英语必修	3.0	80.2
高等数学A上	专业必修	5.0	98.6
2018-2019-2学期			
分析化学C	个性课程	1.5	92.6
分析化学实验C	个性课程	1.0	91
基础生物学	个性课程	3.0	90.07
基础生物学实验	个性课程	1.0	优秀
野外实习	个性课程	1.0	优秀
思想道德修养与法律基础	通识必修	2.5	90.8
交响音乐赏析(GX)	通识选修	2.0	94
社交礼仪(GX)	通识选修	1.5	93
大学英语D2	英语必修	2.0	76
大学物理B	专业必修	5.0	100
高等数学A下	专业必修	5.0	86
2019-2020-1学期			
电工电子实习A	实践课	2.0	良好
初级柔力球	体育必修	1.0	94.1
计算机科学导论	通识必修	2.0	92.3
解读俄罗斯(GX)	通识选修	1.5	96.6
行政管理(GX)	通识选修	2.0	94.4
大学英语D3	英语必修	2.0	75.3
电路原理B	专业必修	3.5	83.6
概率论与数理统计B	专业必修	3.0	81
数据结构C	专业必修	4.0	88
数字逻辑	专业必修	3.0	85.5
物理实验B	专业必修	1.0	91
线性代数	专业必修	2.5	93.1
JAVA语言程序设计D	专业选修	2.5	84.7
面向对象与多线程综合实验	专业选修	1.0	90
2019-2020-2学期			
计算机专业教育与创新创业	实践课	1.0	优秀
计算机组成原理课程设计	实践课	1.0	优秀
初级乒乓球	体育必修	1.0	88.9
马克思主义基本原理	通识必修	2.5	92
毛泽东思想和中国特色社会主义理论体系概论	通识必修	4.5	88
国家公务员制度(GX)	通识选修	2.0	94.1
外国文学概要(GX)	通识选修	2.0	95.3
美国历史与文化	英语必修	2.0	91.5
计算机基础与编程综合实验	专业必修	1.0	95
计算机数值分析	专业必修	2.5	81
计算机网络C	专业必修	2.5	90.2
计算机组成与系统结构	专业必修	4.5	96
离散结构	专业必修	3.5	100
算法设计与分析B	专业必修	2.0	86.4
面向对象程序设计C	专业选修	2.5	93.5
2020-2021-1学期			
编译原理课程设计	实践课	1.0	良好
操作系统课程设计	实践课	1.0	良好
创新创业指导B	实践课	2.0	优秀
初级羽毛球	体育必修	1.0	90.8
高级语言程序设计A	通识必修	3.5	94.8
编译原理	专业必修	3.5	88.6
操作系统B	专业必修	3.5	91.8
计算机网络综合实验	专业必修	1.0	90.7
数据库系统原理B	专业必修	3.5	87.8
数据库系统综合实验	专业必修	1.0	良好
汇编语言程序设计C	专业选修	2.5	81.4
软件工程B	专业选修	2.5	91.5
2020-2021-2学期			
毕业实习	实践课	3.0	优秀
计算机系统能力实训	实践课	4.0	优秀
嵌入式系统应用A	专业必修	2.5	64.2
数据结构与算法综合实验	专业必修	1.0	优秀
UML建模技术	专业选修	2.0	90.2
人工智能概论B	专业选修	2.0	89.4
软件测试B	专业选修	2.0	89.35
以　下　空　白			

学院（盖章）：　　审核人：　　打印时间：2021-09-13

武汉理工大学计算机与人工智能学院计算机科学与技术专业排名证明

兹证明我院计算机科学与技术专业 2018 级学生：李程远，班级：计算机 1804，学号：0121814660102，在校学习期间第 1 至第 6 学期必修课总平均学分绩点：4.307，在 2018 级计算机科学与技术专业 257 人中排名第 4 名，排名百分比为 1.56 %。

特此证明！

武汉理工大学计算机与人工智能学院教学办

2021 年 9 月 15 日

国家奖学金荣誉证书

编号：BZK201912855

李程远 同学荣获 2018 至 2019 学年度本专科生国家奖学金，

特颁此证。

中华人民共和国教育部

2019年12月12日

荣誉证书

李程远同学：

荣获2019—2020学年武汉理工大学“学校奖学金三等奖”。

特发此证，以资鼓励。

武汉理工大学

二〇二〇年十一月

荣 誉 证 书

李程远

在 2018-2019 年度共青团工作先进个人评选中被评为 勤奋好学 先进个人，授予 校优秀团员 荣誉称号。

特发此证，以资鼓励。

共青团武汉理工大学委员会

2019年5月

中国共产主义青年团 武汉理工大学委员会

荣 誉 证 书

李程远：

在 2019-2020 年度共青团工作先进个人评选中被评为 勤奋好学 先进个人，授予 校优秀共青团员 荣誉称号。

特发此证，以资鼓励。

共青团武汉理工大学委员会

二〇二〇年

荣誉证书

李程远同学：

在2018—2019学年中，德智体美劳全面发展，被评为武汉理工大学“校三好学生”。

特发此证，以资鼓励。

武汉理工大学

二〇一九年十一月

荣誉证书

李程远同学：

在2019—2020学年中，德智体美劳全面发展，被评为武汉理工大学“校三好学生”。

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武汉理工大学

二〇二〇年十一月

2021

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Chengyuan Li
Hanjun Luo
Haoran Zhu

With Faculty Advisor

Congjun Rao

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代表 武汉理工大学 在

2021 年“中国高校计算机大赛 – 团体程序设计天梯赛”

全国总决赛“华山论剑”组中荣获

团队二等奖

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主办单位

证书编号：CCCC2021GPLT06937

组委会主任签名：

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2020

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Hanjun Luo

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三等奖

特发此证，以资鼓励。

全国高等学校计算机教学研究会　　中国计算机学会系统软件专委会　　中国计算机学会软件工程专委会　　绿色计算产业联盟

二零二零年十一月

中国高校计算机大赛

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证 书

2021 团 体 程 序 设 计 天 梯 赛

李程远 同学

作为 WUT_2 队成员之一

代表 武汉理工大学 在

2021 年“中国高校计算机大赛 – 团体程序设计天梯赛”

湖北省“华山论剑”组中荣获

团队一等奖

特发此证 以资鼓励

主办单位

证书编号：CCCC2021GPLT19314

组委会主任签名：

全国高等学校计算机教育研究会

2021 年 5 月 6 日

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证 书

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李程远 同学

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全国总决赛中勇夺

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组委会主任签名：

主办单位

全国高等学校计算机教育研究会

2021 年 5 月 6 日

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武汉理工大学李程远：

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证件号码：342423200009216557

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中国软件行业协会

中国电子学会

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中国半导体行业协会

2020 年 10 月 17 日

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获奖证书

武汉理工大学李程远：

荣获第十二届蓝桥杯全国软件和信息技术专业人才大赛湖北赛区 C/C++程序设计大学 A 组三等奖。

特发此证，以资鼓励。

证书编号：011201502

证件号码：342423200009216557

工业和信息化部人才交流中心

蓝桥杯全国软件和信息技术专业人才大赛组织委员会

中国软件行业协会

中国电子商会

中国电子学会

中国半导体行业协会

2021 年 5 月 31 日

中国高校计算机大赛

证 书

2021 网络技术挑战赛

华中赛区

三等奖

参赛学校：武汉理工大学

参赛团队：风中追风

组　　别：本科

指导教师：何凡

参赛队员（排名不分先后）：

张仕华　陈巨星　李晓雨
曹　浪　李程远

证书编号：CCCC2021NTC0306

组委会主任：

主办单位

全国高等学校计算机教育研究会

2021年8月

—NCCCU—

荣誉证书

— HONORARY CREDENTIAL —

武汉理工大学　　李程远

参加全国高等学校计算机教育研究会主办的“2020年全国高校计算机能力挑战赛”，成绩优异，荣获程序设计赛（C++）华中区域

三 等 奖

特发此证，以资鼓励！

全国高等学校计算机教育研究会
全国高校计算机能力挑战赛组委会
二〇二一年六月

编号：N2656984037060

荣譽證書

祝贺李程远同学在二零一九年武汉理工大学第七届“新生杯”数学建模比赛中荣获“一等奖”

特发此证，以资鼓励。

武汉理工大学数学建模协会

二零一九年十二月

荣誉证书

HONORARY CREDENTIAL

李程远 同学：

你在武汉理工大学第十七届“理苑星空”大学生科技文化节之校第十届数学课程学习知识竞赛中成绩优异，荣获 一 等奖。

特发此证，以资鼓励。

武汉理工大学第十九届“理苑星空”
大学生科技文化节组委会
（理学院代章）
二〇二〇年六月

证书

2021年中国高校计算机大赛-团体程序设计天梯赛校赛

获奖等级：一等奖

获奖学生：李程远

武汉理工大学教务处

二〇二一年三月廿七日

计算机程序设计能力考试
Programming Ability Test

成绩证书（甲级）
Official Score Certificate (Advanced Level)

姓名（Name） 李程远

性别（Gender） 男

身份证号码（Official ID No.）
342423200009216557

准考证号码（Registration ID No.）
A2099202109110469

PAT
计算机程序设计能力考试
Programming Ability Test

兹证明此证书持有者于 2021 年 9 月 11 日参加了程序设计能力考试（PAT）之甲级，成绩如下：

This is to certify that the bearer of this certificate completed the Advanced Level of Programming Ability Test (PAT) on September 11, 2021, with the following scores:

段位 Division	总分/满分 Total/Full Mark	排名/本场考生人数 Rank/Total Number of Testees
王者	91 / 100	585 / 2189

证书编号（Certificate No.）：A210911218920991954
查询网站（Website）：https://www.patest.cn/certificates

全国高等学校教学研究中心
CHINA RESEARCH CENTER FOR TEACHING AND LEARNING IN UNIVERSITIES AND COLLEGES

浙江大学
ZHEJIANG UNIVERSITY

Soft Computing

Spread and classification of the Asian giant hornets based on improved logistics model and cost-sensitive random forest model

--Manuscript Draft--

Manuscript Number:	
Full Title:	Spread and classification of the Asian giant hornets based on improved logistics model and cost-sensitive random forest model
Article Type:	Original Research
Keywords:	Asian giant hornet; Improved grey prediction model; Logistic model; Cost-sensitive RF model; LDA model
Corresponding Author:	Congjun Rao Wuhan University of Technology CHINA
Corresponding Author Secondary Information:	
Corresponding Author's Institution:	Wuhan University of Technology
First Author:	Chengyuan Li
First Author Secondary Information:	
Order of Authors:	Chengyuan Li
	Haoran Zhu
	Hanjun Luo
	Congjun Rao
Funding Information:	National Natural Science Foundation of China (Nos. 71671135, 72071150) — Prof. Congjun Rao
Abstract:	It is of great significance to systematically study the spread rules and classification model of the Asian giant hornets for the correct identification of the Asian giant hornets and the use of limited resources to investigate and control its wanton spread. In this paper, we established a prediction model of hornets spread from two perspectives of spatial and quantitative distribution, and solved the classification problem of Asian giant hornets in unbalanced data sets by establishing a cost-sensitive random forest (RF) model. Firstly, in order to solve the problem of hornet's spread rules, this paper considered the spread range and the spread quantity of hornets. On the one hand, an improved grey prediction model was established to systematically predict the latitude and longitude of hornets with time. On the other hand, through the establishment of logistic model, we obtained the law of the number of hornets changing with time. Secondly, in order to solve the problem of hornets classification and priority survey decision, we established a cost-sensitive RF model. Through the steps of feature extraction, conversion of unbalanced data set and training data set, the hornets dichotomy model was established, and the recognition accuracy of the model was 0.918. Thirdly, to solve the problems of reports update and pest eradication, we established an improved Logistic model by adding the human control factors and cycle parameters on the basis of the regularity of spread quantity, and obtained the judgment conditions of report update frequency and elimination of pests and disease. Finally, we test the goodness of fit for the results of each model, and the test results show that the models established in this paper are feasible and reasonable. The result in this paper provides a good theoretical support and decision-making reference for the government to make effective investigation and control of hornets with limited resources.
Section/Category:	Application of soft computing

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Spread and classification of the Asian giant hornets based on improved logistics model and cost-sensitive random forest model

Chengyuan Li [b], Haoran Zhu [a], Hanjun Luo [a], Congjun Rao [a, *]

[a] *School of Science, Wuhan University of Technology, Wuhan 430070, P. R. China*

[b] *School of Computer Science and Technology, Wuhan University of Technology, Wuhan 430070, P. R. China*

Abstract It is of great significance to systematically study the spread rules and classification model of the Asian giant hornets for the correct identification of the Asian giant hornets and the use of limited resources to investigate and control its wanton spread. In this paper, we established a prediction model of hornets spread from two perspectives of spatial and quantitative distribution, and solved the classification problem of Asian giant hornets in unbalanced data sets by establishing a cost-sensitive random forest (RF) model. Firstly, in order to solve the problem of hornet's spread rules, this paper considered the spread range and the spread quantity of hornets. On the one hand, an improved grey prediction model was established to systematically predict the latitude and longitude of hornets with time. On the other hand, through the establishment of logistic model, we obtained the law of the number of hornets changing with time. Secondly, in order to solve the problem of hornets classification and priority survey decision, we established a cost-sensitive RF model. Through the steps of feature extraction, conversion of unbalanced data set and training data set, the hornets dichotomy model was established, and the recognition accuracy of the model was 0.918. Thirdly, to solve the problems of reports update and pest eradication, we established an improved Logistic model by adding the human control factors and cycle parameters on the basis of the regularity of spread quantity, and obtained the judgment conditions of report update frequency and elimination of pests and disease. Finally, we test the goodness of fit for the results of each model, and the test results show that the models established in this paper are feasible and reasonable. The result in this paper provides a good theoretical support and decision-making reference for the government to make effective investigation and control of hornets with limited resources.

Keywords Asian giant hornet, Improved grey prediction model, Logistic model, Cost-sensitive RF model, LDA model

1 Introduction

In September 2019, a colony of Vespa mandarinia (also known as the Asian giant hornet) was discovered on Vancouver Island in British Columbia, Canada. Although the nest was soon destroyed, the news of this event spread rapidly throughout the area. Since then, several confirmed sightings of the pest have occurred in neighboring Washington State, as well as a number of mistaken sightings.

The Asian hornet in Washington state is a species of wasp from China that is four times larger than native bees in Britain and France, making it the largest wasp species in the world. It is characterized by a striking color with a yellow head, black chest and yellow and black or brown striped abdomen. Because this kind of hornets is the predator of European bees, they invade and destroy their nests, a small number of hornets can destroy the whole European bee colony in a short time, resulting in a serious decline in honey production and being regarded as an alien invasive species locally. Moreover, this kind of hornets is also aggressive, causing people to be stung, and people with allergic constitution can also cause death. The presence of the Asian hornet has brought great anxiety to local residents.

* Corresponding author: Congjun Rao, cjrao@163.com

The life cycle of this hornet is similar to many other wasps. Fertilized queens appear in the spring and start a new colony. In autumn, the new queens leave the nests to spend winter in the soil, waiting for the arrival of spring. A new queen builds her nest in the range of 30 kilometers. The presence of Vespa mandarinia can cause a lot of anxiety due to the potentially serious impact on local honeybees. Washington State has established a hotline and website for people to report sightings of these hornets. Based on these reports from the public, the state must decide how to prioritize its limited resources for further follow-up investigations. Although some reports have been identified as the Asian giant hornets, many other sighted insects have proven to be other types of insects.

Therefore, the identification of these hornets and the effective follow-up investigation become an important issue. Especially, it is of great importance to study on the prediction of hornets population propagation on the number and spatial distribution and on the classification of hornets based on information such as images and text. Based on the above research background, this paper aims to study the spread and classification of the Asian giant hornets, which will provide a good theoretical support and decision-making reference for the government to effectively investigate and control hornets under the limited resources.

2 Literature review

2.1 Research status of hornet's spread and prediction models

The Asian giant hornet is an insect and an invasive species. All kinds of harm caused by alien invasive species is a global problem that human society needs to solve urgently. A large number of experts and scholars have made systematic studies on it from various aspects, and have obtained many results with practical reference value in the aspects of insect spread and prediction model.

The spatial and temporal dynamics of insect population include the change of population quantity and spatial distribution. However, the research on insect population dynamics mainly focuses on the quantitative dynamics, and there are few studies on the spatial change, especially the spatiotemporal change. Wu (1963), Zhao and Zhou (1984) made some careful studies, but such models did not consider the process of population life and death, as well as the characteristics of distribution pattern in the process of population change. On this basis, Zhang and Gu (2001) deduced a nonlinear partial differential equation model describing the spatial and temporal distribution of insect population according to the density constraint principle, and obtained the number of insects at each point in the field at each moment, as well as the spatial distribution type and clustering at this moment. Their model has a better prediction of the insect populations in the change of the two aspects of quantity and space with high accuracy, but they only considered three initial distribution hypothesis, i.e., the binomial distribution, Poisson distribution and negative binomial distribution. However, the initial distribution of the actual insect population is more complex and does not always fit one of the above categories well.

Grey system analysis can predict the future reliably based on a small amount of historical information. Therefore, the grey prediction model is widely used to solve the propagation prediction problems with very little given data. Liu and Liu (1990) worked out the life table of the wasp, understood the change law that affected its population, and predicted it with the analysis theory of development dynamics in the grey system. This method overcomes the disadvantages of traditional modeling, such as large amount of data, typical distribution and large amount of calculation, and puts forward a new way for forest disease and insect prediction in the future. Wang et al. (1992) used grey system theory and method to explore the possibility of cotton bollworm prediction. Lin and Bureau (2018) used GM(1,1) based residual correction model to study the relationship between forest diseases and insect pests and cutting amount, and predicted their future quantity trend, so as to provide support for nursery stock and forest pest management. The classic grey prediction model is used on the premise that the time interval between the data is equal, but in

reality, there are quite a lot of data with unequal time isolation need us to forecast. Therefore, the problem of insect propagation prediction with unequal time interval is worth our research. Therefore, we established an improved grey prediction model. Taking the correct reporting time of the Asian giant hornets sightings in Washington state as an example, we processed the data with unequal time intervals, and successfully used the improved grey model system to predict the range of latitude and longitude of hornets occurrence over time (Deng 1997; Xie 2020).

In recent years, the cellular automata model has also been applied to study the propagation model of hornets. Lin and Zhou (2009) used mathematical model combined with spatial autocorrelation function to establish a spatio-temporal prediction model as a conversion rule, and used the cellular automata model to simulate the propagation of P. rubifolia on the platform of geographic information system. The predicted results provided the results of the density distribution of P. rubifolia in Kunyu Mountain, which provided convenience for prevention and control decisions. Pan et al. (2021) also studied the propagation of Asian giant hornets in Washington State by establishing a cellular automata model. Meanwhile, the paper constructed a corresponding diffusion model and carried out a detailed study on the distribution and diffusion of wasps in space. Cellular automata adopts a typical "bottom-up" modeling method, which conforms to the formation law of complex systems, and is also the thinking mode adopted by most complex systems research. It is a complexity research method advocated by complex science. At the same time, the cellular automata has the advantages of universality, flexibility and openness. However, in the standard cellular automata, the cells have a regular uniform shape and are arranged regularly in the cellular space. But in the real world there are few such regular states. Moreover, the cellular automata has some significant shortcomings in the determination of the geometric shape of the cellular space, the definition of the cellular neighborhood and the determination of the cellular state update rules. It is worth noting that the deficiencies of the above standard cellular automata have been gradually improved, so most of the application models of cellular automata are no longer standard cellular automata models. However, the improvement of the standard cellular automata needs to be further improved in order to improve the simulation and application ability of cellular automata to the real world (Wang et al. 2018; Taichiro and Takuma 2021; Pereira et al. 2021).

Matrix models are the main mathematical tools for simulating population dynamics and are widely used in the study and discussion of insect population propagation and prediction. Since Leslie (1945) introduced matrix model into the study of biological population dynamics, this method has been widely used and developed. Pang et al. (1980) improved the matrix model and solved the matrix model problem of age groups in different periods, so that the Leslie matrix model was applicable to insect population. However, Pang's model required that the dimension of the matrix was unchanged, so Xu et al. (1981) proposed the variable dimension matrix model, which better solved the problem under the condition of variable temperature. Modeling population dynamics problems. Variable dimension matrix model is a temperature-dependent model based on individual ecological data, which has some problems. Qin and Li (1995) used the improved model to simulate the population dynamics of wheat moth and other species, aiming at the situation that the matrix dimension of Xu's model might be larger than the period, and there would be delay in the simulation process, so that the matrix dimension was consistent with the development period. Zhang et al. (1994) improved the simulation method of insect population dynamics based on the fact that Xu's model did not involve the influence of developmental rate on population dynamics, considering the influence of temperature, nutritional conditions and genetic factors. Huang and Ye (1995) proposed an improved variable dimension matrix model to simulate insect population dynamics, which took the development period as the dimension, changed the dimension by means of decomposition and synthesis, and took into account the development differences between individuals. By reviewing on insect populations in recent decades based on the research of the simulation model of development, and fully

considering the different insect population development characteristics, Zhao and Shen (1999) established a general development model with developmental stages and age structure. Zhao's model can be used in the simulation with different number, and the attitude of the insect growth course of the number of different age groups, with decoupling union under the condition of the weather model, can be used for the prevention and control of insect population prediction of the dynamic and different strategies of risk analysis. The results showed that the model has good performance through the simulation of corn borer life table data.

By studying the population ecological dynamics system and infectious disease dynamics system in a wider range of dimensions and building the corresponding differential equation model, we can quantitatively study the propagation and prediction of insect population from the perspective of biological mathematics. There are many researches using these two systems and many important theoretical achievements have been obtained. According to the actual situation in the ecosystem, the expression of some coefficients is changed, and the time-varying coefficients are considered, that is, some coefficients change with the change of time or season, so that the newly constructed model is more in line with the actual situation (Hadeler 2011; Mummert 2013). Delay differential equation modeling is also well generalized in biological invasion field (Tchuenche and Nwagwo 2009; Wang and Zhang 2013). In recent years, the impulse differential equation method is more practical when it describes some parameters that do not change continuously, but change periodically or change rapidly at a non-fixed time (Muntaser et al. 2013; Dong 2010). In addition, it was found that species of different ages showed different correlation coefficients such as transmission rate, mortality rate and removal rate, which led to the emergence of a partial differential equation model considering the age structure of species. Li et al. (2019) studied an age structure model of mosquito plasticity with a strong biological background, proved the existence of solutions of a class of generalized systems, the principle of comparison, the existence of optimal harvest and optimal control, and established the necessary optimal conditions. Yang et al. (2021) mainly discussed a kind of multi-stage age structure population model based on the biological background of insect control. Taking the European vine moth as an example, its growth cycle is divided into four stages: egg, larva, pupa and adult. The mathematical model of this kind of insect is a set of partial differential equations with non-local boundary conditions.

Logistic equation is a classic ecological model, which points out the basic law of population growth in limited space. It has been widely used in the research field of population ecology to describe the spatiotemporal dynamic relationship of biological population. In recent years, the study of Logistic model mainly focuses on the modification of the equation, parameter fitting and the optimal capture strategy. Shen (1985) discussed the modification of the Logistic equation, derived the Logisic-R equation and Logisic-P equation, and used the latter to fit the field observation data of population density dynamics of Aphid japonicus with SD curve, and achieved good results. There are also many researches on parameter estimation methods of Logistic equation, but most of these methods assume that the three parameters are independent of each other and perform data fitting analysis on Logistic equation. Tang et al. (1996) pointed out that there were common errors in these parameter estimation methods and proposed specific correction methods. In addition, Bo et al. (2004) discussed the problem of optimal capture strategy for stable bounded Logistic equation, chose the limit mean of maximum persistent return per unit time as the management objective, and obtained the optimal population level. Taking Logistic equation with almost periodic coefficient as an example, it is shown that this result not only generalizes the harvest problem of the classical Logistic equation about autonomy, but also generalizes the result of the harvest problem of the Logistic equation about period. Yang et al. (2019a) studied the optimal harvesting strategy of Logistic population model under a kind of random fluctuation environment, and concluded that in order to obtain the optimal harvesting strategy, the harvesting workload and noise intensity must be reduced. Kumar and Rajeev (2020) discussed a spatial fractional diffusion logistic population model with Caputo fractional

derivative and density dependent diffusivity. The numerical solution of the problem was obtained using the finite difference scheme. The effects of density-dependent diffusion rate and spatial fractional derivative on population density and spread front (moving boundary) are analyzed. In the study of Logistic model by domestic and foreign scholars, few of them take human control factors and periodic parameters into comprehensive consideration. Therefore, in view of this problem, we took the Asian bumble bee of Washington state as an example, firstly established the classic Logistic model to predict the population propagation law of Asian bumble bee, and then established the improved Logistic model with the addition of human control factors and periodic parameters to forecast and analyze the population situation of bumble bee. This will enable the Washington State government to develop effective recommendations and protective measures against the Asian bumblebee invasion (Sakanoue 2007).

2.2 Research status of hornet's classification models

Entomological taxonomy mainly studies the identification of insects and their genealogy, involving the identification, nomenclature, classification of insects as well as the phylogenetic relationship and the way of development among different stages. In recent years, numerical taxonomy, cladistics, chemical taxonomy, cellular taxonomy and molecular taxonomy of insects have been formed. Scholars have made more detailed research and discussion on insect classification methods and models.

Mathematical morphology classification is the most basic method in insect classification. This method is based on the principle that the morphological structure is consistent with the function, and based on the external morphology, internal structure and the size proportion of each part of insects, which is often used to classify and identify the insects. Shen et al. (2003) proposed the idea of using mathematical morphology to extract the features of insect images, discussed the relationship between the mathematical morphological features of insects and insect taxonomy and evolution, and developed the automatic recognition software Bug Visux for insect images. The system includes the collection of insect digital image, the segmentation of insect image, the feature extraction and classification of image. On this basis, Zhao et al. studied the phylogenetic relationship of insect groups with the same taxonomic level from the perspective of mathematical morphology, and upgraded the system Bug Visux (Zhao 2002; Zhao et al. 2003a; Zhao et al. 2003b; Zhao et al. 2003c). However, Shen and Zhao's research requires that the production of insect specimens must be very standardized, and the acquisition of insect images must be completed under specific conditions in the laboratory, so the anti-noise ability is poor. In order to solve the above problems, Cai et al. (2016), Cai et al. (2019) carried out geometric morphometry analysis on the forewings of 10 species of moths, discussed the feasibility of using geometric morphometry to realize digital classification of moths, and provided new morphological basis for gradually realizing digital classification of moths.

The ethological classification of insects is a method to identify and classify insects according to some behavioral characteristics, such as sexual behavior, pre-mating behavior, mating posture, circadian rhythm, feeding behavior and songs. Lian and Li (2002) studied the three common chirps of three species of crickets and found that these three types of chirps have differences in frequency domain and time domain characteristics, but they also have some similar features, which are of great significance in the classification of song worms. Xiang et al. (2016) reviewed the application of sex pheromone courtship behavior, song courtship behavior, flash courtship behavior and gift courtship behavior in taxonomy, in order to help distinguish the related species of specific groups, which is of great significance for the identification of biological species. The method of insect behavioral classification and identification is relatively simple, but the accuracy is not high, so it can assist other insect classification methods to be used in the classification and identification of insects (Huetteroth and Pauls 2019; Takahiro and Takema 2020).

The traditional species classification and identification mainly rely on morphological observation, but for some species with great similarity in appearance characteristics, relying solely on morphological

identification often leads to some errors. In this case, more accurate genetic methods are needed to confirm. Cytogenetic classification is mainly used to identify insect species through comparative analysis of the number and morphology of chromosomes, chromosomal grouping type, sex chromosome position, C-banding type and its components, total content of heterochromatin, and meiotic behavior pattern of chromosomes. Wei (2004) studied the karyotypes and C-banding patterns of several Orthoptera insects from the perspective of cytogenetics, and classified them. Due to the characteristics of convenience, economy and accuracy of cytogenetic classification methods, the use of cytogenetic methods for the classification and identification of the species of Tryphagidae has greatly improved the accuracy of the classification and identification of the species (Alevi et al. 2016; Wang MX and Liu Q 2021).

The main applications of biochemistry in insect classification include isoenzyme electrophoresis and gas chromatography for the determination of epidermal hydrocarbons. Zhang et al. (1999) conducted a comparative study on the esterase isozymes of five species of ladybird beetles, and the results showed that there were significant differences in the enzyme spectra among genera, while there were some commonalities within genera. Cui et al. (1992) analyzed the cortical hydrocarbons of Anopheles from different regions and seasons in Hainan Province by gas chromatography, and concluded that they belonged to the same species. Biochemical methods study the differences of biochemical characteristics in the process of insect evolution at the cellular level, and combine them with corresponding morphological characteristics organically. Therefore, they provide a reliable theoretical basis for classical morphological taxonomy. Especially in the last ten years, biochemical taxonomy has made some achievements. Compared with the classical morphology, its data achievements are relatively poor, but it has a great prospect and should not be underestimated (Urszulad et al. 2020; Toprak and Musselman 2021).

Molecular classification is also widely used in insect classification. Xie et al. (2001), Ren et al. (2012) systematically introduced the molecular biology techniques commonly used in insect classification and identification, including molecular hybridization technology, PCR technology, RFLP technology, RAPD technology, SSCP and DSCP technology, DNA barcode, and combined with their own work, elaborated the application prospect of molecular biology insect identification methods in plant protection. In recent years, DNA barcoding technology has been widely concerned by scholars because of its easy operation, high accuracy and rapid identification of a large number of samples at one time. Gisele et al. (2017) used insect DNA barcoding sequence detection and identification technology to classify and identify insects, analyze population differentiation, discover hidden species and new species. At present, some gene sequences in DNA barcodes have been widely used in insect molecular biology research. With the development of information technology in the new era, the detection technology of insect DNA barcodes has great influence on insect taxonomy, which has important theoretical and practical significance (Chen et al. 2020).

In recent years, the research of insect classification and recognition based on image has developed gradually. Image recognition based on machine learning generally consists of two steps: feature extraction and classifier training (Rao et al. 2021; Wen et al. 2020). The features commonly used in insect recognition can be divided into global features and local features. Global features include color, texture, shape, and so on. Local characteristics mainly include HOG, SIFT, LBP, PCBR and so on. The classifiers commonly used in machine learning for insect features include decision tree, KNN, neural network, support vector machine, K-means, and so on. Luo (2011) applied support vector machine to entomological research, which solved the problems existing in traditional entomological taxonomy. Liu (2019) realized automatic insect detection in different scenarios, effectively verifying the feasibility of target detection technology based on deep learning in the field of insect surveillance. The improved lightweight detection model can provide core identification technology for the establishment of the Internet of Things for forest insect surveillance. In order to improve the recognition rate of insect images taken in the natural environment,

Xie et al. (2018) adopted the derived algorithm of artificial neural network, namely Faster R-CNN algorithm, to recognize butterfly images taken in the natural environment, and achieved a high recognition rate. In the case of a small number of samples and limited species of agricultural pests, these methods rely on manual design of features, training and testing of classifiers, and can generally achieve good identification accuracy.

However, in the complex field environment and a wide variety of insect species, the robustness and generalization ability of traditional machine learning methods are relatively poor, and it is still difficult to be applied in practical production. In recent years, deep learning technology has made great achievements in the field of image recognition. Among the representative ones, Ding and Taylor (2016) combined convolutional neural network and sliding window to detect, recognize and count the insect images on the sex attractant adhesive plate. By adopting the image processing technology of deep learning, Fu (2020) proposed a set of image segmentation and image recognition methods based on convolutional neural network, and completed the automatic recognition and counting of crop insect images collected in the field. Mo et al. (2014) collected samples' near-infrared spectral data by integrating spherical reflection mode, pretreated the original spectrum by Savitzky-Golay first-order differential method, and established a qualitative discrimination model between spectral data and honey authenticity by using random forest method. Compared with other methods, this method has the advantages of simple operation, fast speed, high efficiency, no pollution, low cost and no complicated pretreatment. In recent years, many classic algorithms related to machine learning have been used in insect classification and recognition, but the application of ensemble learning (such as random forest algorithm) algorithm in insect classification is rarely mentioned in papers, and there are even fewer studies on the case that the training data set is unbalanced. Therefore, we established a cost-sensitive random forest model for the classification of insects in unbalanced data sets (taking the Asian bumble bee of Washington State as an example), which effectively improved the classification accuracy in unbalanced data sets (Yang et al. 2018; Yang et al. 2019b; Mercadier and Lardy 2019; Rao et al. 2020b).

3 Prediction Model of Spread Rules

3.1 Data preprocessing

This problem requires us to predict the spread of the Vespa mandarinia over time and provide the level of prediction precision. Therefore, we need to make a study of the Vespa mandarinia which is not mistaken in the data. After data screening, we found a total of 14 Asian giant hornets that appeared at different times.

3.2 Prediction of spread range

To predict the spread rang, it is obviously necessary to predict the latitude and longitude changes of the hornets. First, we fit the latitude and longitude of the locations of the 14 Asian giant hornets and combine the map given in the question to obtain the approximate spread range. Second, we establish an improved grey forecasting model to systematically predict the range of changes in the latitude and longitude of the reported hornets over time. Finally, we make a comparison and draw the conclusion about the spread range of the hornets.

3.2.1 Data fitting for longitude and latitude

In order to predict the latitude and longitude of the Asian giant hornet's range of activity, we fit the data of the latitude and longitude. Taking the time when the hornet was first found September 19, 2019 as the starting point, we arrange continuously the time when the hornet was found to fit the latitude and longitude. The purpose of fitting is to find a function which can make the data points corresponding to the known longitude and latitude fall on or near the function image as much as possible, and minimize the sum of

squared errors from the point to the function, that is

$$SSE=\sum_{i=1}^{14}(y_i-y_i^{'})^2\ , \tag{1}$$

where y_i represents the longitude or latitude information of the data point, $y_i^{'}$ represents the value of the longitude or latitude of the data point on the fitting function, and SSE represents the sum of squared errors.

We use the Matlab toolbox to perform function fitting and put the obtained latitude and longitude change trend graph into the map. The results are shown in the Figure 1 as follows.

Fig. 1 The results of fitting about Asian giant hornets' spread range

Note: The red circle represents the distribution of Asian giant hornets in the data, and the blue circle represents the distribution of Asian giant hornets predicted by fitting.

As can be seen from Figure 1, Asian giant hornets 's spread direction is roughly from northwest to southeast, that is, from Vancouver to Washington. The range of latitude and longitude is approximately 48.75° N to 49.20° N, and 121.00° W to 124.50° W.

3.2.2 Improved grey prediction model

Due to the small amount of data (only the 14 correct data reports) in this topic, and in order to make the prediction effect more ideal, we establish an improved grey prediction model in this section. Through the equal spacing processing, smoothness test, and finally the improved grey prediction model GM(1,1) model is used to predict the longitude and latitude of Asian giant hornets' activities with the change of time(Deng 1997; Xie 2020).

Step 1: Processing of non-equidistant data

Since the time of data in this problem is non-equidistant, and the grey prediction model should be carried out in the case of equidistant, it is necessary to carry out equidistant processing on the original data.

Assume that the original data sequence is

$$x^{(0)}(k_i)=\{x^{(0)}(k_1),x^{(0)}(k_2),...,x^{(0)}(k_n)\}\ , \tag{2}$$

where n =14 (the same below), and $x^{(0)}(k_i)$ represents the longitude and latitude sequence of 14 Asian giant hornets.

The time interval between each reporting date and the first reporting date (the time the first Asian giant hornet was found and reported) is set as

$$K_i = k_i - k_1, i = 1, 2, ..., n. \tag{3}$$

Then the average time interval is

$$\Delta k_0 = \frac{k_n - k_1}{n-1}. \tag{4}$$

The coefficient of unit time difference between k_i and Δk_0 is

$$u(k_i) = \frac{k_i - (i-1)\Delta k_0}{\Delta k_0}. \tag{5}$$

Find the total difference of each period

$$\Delta x^{(0)}(k_i) = u(k_i)\{x^{(0)}(k_i) - x^{(0)}(k_{i-1})\}. \tag{6}$$

Calculate the grey value of equidistant points

$$Z^{(0)}(k_i) = x^{(0)}(k_i) - \Delta x^{(0)}(k_i). \tag{7}$$

Construct the equally spaced sequences

$$Z^{(0)}(k_i) = \{Z^{(0)}(k_1), Z^{(0)}(k_2), ..., Z^{(0)}(k_n)\}. \tag{8}$$

Make a summation of $Z^{(0)}(k_i)$ to generate (1-Ago) to generate sequence

$$Z^{(1)}(k_i) = \{Z^{(1)}(k_1), Z^{(1)}(k_2), ..., Z^{(1)}(k_n)\}, \tag{9}$$

where $Z^{(1)}(k_i) = \sum_{m=1}^{k_i} Z^{(0)}(m), k_i = 1, 2, ..., n$.

Step 2: Smoothness test

To ensure the feasibility of the grey model, the smoothness test of $X^{(0)}$ is required before the application of the model. The formula is as follows:

$$\rho(k_i) = \frac{Z^{(0)}(k_i)}{Z^{(0)}(k_i - 1)}, k_i = 2, 3, ..., n. \tag{10}$$

If $\rho(k_i) < 0.5$ is satisfied, then the GM(1,1) model can be established to predict. Otherwise, the data needs to be translated, and the formula is as follows:

$$y^{(0)}(k_i) = Z^{(0)}(k_i) + c. \tag{11}$$

After the test, $\rho(k_i) = 0.23 < 0.5$, the test passed.

Step 3: Establishment of GM(1,1) model

The original form of the GM(1,1) model is as follows:

$$Z^{(0)}(k_i)+aZ^{(1)}(k_i)=b\,. \tag{12}$$

We set

$$E^{(1)}(k_i)=\frac{1}{2}\left(Z^{(1)}(k_i)+Z^{(1)}(k_i-1)\right), \tag{13}$$

where $E^{(1)}(k_i)=(E^{(1)}(2),E^{(1)}(3),...,E^{(1)}(14))$ is the generate sequence by adjacent mean values of $X^{(1)}(k_i)$. Therefore, the GM(1,1) model is transformed into the following basic form:

$$Z^{(0)}(k_i)+aE^{(1)}(k_i)=b\,. \tag{14}$$

We set

$$\hat{\alpha}=(a,b)^T\,, \tag{15}$$

$$Y=(Z^{(0)}(2),Z^{(0)}(3),...,Z^{(0)}(n))^T\,, \tag{16}$$

$$B=\begin{bmatrix}-E^{(1)}(2) & 1\\ -E^{(1)}(3) & 1\\ ... & ...\\ -Z^{(1)}(n) & 1\end{bmatrix}, \tag{17}$$

then $\hat{\alpha}=(B^TB)^{-1}B^TY$ is the estimated parameter sequence of the grey differential equation (12) by the least square method, and the time response sequence of $Z^{(0)}(k_i)+aE^{(1)}(k_i)=b$ can be obtained as follows:

$$\hat{Z}^{(1)}(k_i+1)=(Z^{(1)}(0)-\frac{b}{a})e^{-ak_i}+\frac{b}{a},k_i=1,2,...,n\,. \tag{18}$$

We set $Z^{(1)}(0)=Z^{(0)}(1)$, then we can get:

$$\begin{aligned}\hat{Z}^{(0)}(k_i+1)&=\hat{Z}^{(1)}(k_i+1)-\hat{Z}^{(1)}(k_i)\\&=(1-e^a)(Z^{(0)}(1)-\frac{b}{a})e^{-ak_i},k_i=1,2,...,n\end{aligned} \tag{19}$$

Substitute the longitude and latitude data of 14 hornets into the above model (19) and use the MATLAB software to solve the model, and put the variation trend into the map, then we obtain the results as shown in Figure 2.

It can be found from Figure 2 that the spread direction of Asian giant hornets is also roughly from northwest to southeast, that is, from Vancouver to Washington, with latitude and longitude ranging from 48.60° N to 49.20° N and 121.30° W to 124.80° W.

In conclusion, the spread range of Asian giant hornets over time is as follows: from Vancouver to Washington, the range of latitude and longitude is approximately 48.60° N to 49.20° N, and 121.00° W to 124.80° W.

Fig. 2 The results of grey prediction about Asian giant hornets' spread range

Note: The red circle represents the distribution of Asian giant hornets in the data, and the orange circle represents the grey predicted distribution of Asian giant hornets.

3.3 Prediction of the spread quantity

Since Asian giant hornets belong to a natural population, we assume that their biological evolution is similar to that of other common insects and bees. Therefore, for the spread quantity of Asian giant hornets, we establish a Logistic population growth model to predict it.

3.3.1 Logistic model

According to relevant literature (Sakanoue 2007), hornets in a honeycomb will die around December in winter, leaving only the queen to hibernate. In spring, the queen will come out to build the nest and lay eggs, and only the fertilized queen bees can lay eggs. By August, the number of worker bees is at its peak, about 100, and September is the month when males and queens are born, so we assume that the breeding of Asian giant hornets starts in early March. Because the number of males and queens is already low, and the queen bee is likely to fight off the males, with a 65% chance of failure to fertilize.

Therefore, we take a cycle of a year and assume that the initial number of hornets in a hive is $N(0)=1$, and the maximum number of hornets is $N_{\max}=100$ by the end of August. Assume that the natural growth rate of hornets is a constant *r*, then $r=\frac{\Delta N(t)}{t}=\frac{100}{6}$. Thus, the Logistic growth model is obtained as follows:

$$\begin{cases} \frac{dN}{dt}=rN(1-\frac{N}{N_{\max}}) \\ N(0)=1 \end{cases}, \tag{20}$$

where $N(t)$ represents the number of hornets over time, *r* represents the growth rate of hornets, $N_{\max}$ represents the environmental capacity of hornets, then *rN* represents the growth trend of hornets themselves, and factor $(1-N/N_{\max})$ represents the blocking effect of environment and resources.

The separation of variables method is used to solve the ordinary differential equation (20), and the general solution is obtained as follows:

$$N(t)=\frac{N_{\max}}{1+(N_{\max}-1)e^{-rt}}\ . \tag{21}$$

The life cycle of Asian giant hornet is one year. In early March, the queen bees come out to build nests and lay eggs, by late August, the number of hornets reaches its peak. Because of the climate, until the beginning of December, when winter sets in, the hornet population will decline until it's all dead, and the new queen will find a place to hibernate, and then come out again the following spring. Therefore, according to the above analysis, hornets will gradually decrease with the climate from the beginning of September until they all die, leaving the new queen bee to hibernate. Therefore, for September to December, we establish a new hornet population change model as follows:

$$\begin{cases}\dfrac{dN}{dt}=r'N\\ N(6)=N_{\max}\end{cases}, \tag{22}$$

where r' ($r'<0$) is the mortality rate, and is also the influence factor of climate on hornets. Hornets will be reduced to about 1 to 2 queens within 3 months, which can be simplified as follows:

$$N(t)=\frac{N_{\max}}{e^{6r'}}e^{r't},6<t\le 9\ . \tag{23}$$

In summary, the reproductive law model of a hornet's life cycle in a honeycomb can be expressed by

$$N(t)=\begin{cases}\dfrac{N_{\max}}{1+(N_{\max}-1)e^{-rt}},0\le t\le 6\\ \dfrac{N_{\max}}{e^{6r'}}e^{r't},6<t\le 9\end{cases}\ . \tag{24}$$

3.3.2 Solution of the model

Use the Matlab software to describe the function graph of Model (24), and the curve for the number of hornets changing with time can be obtained, as shown in the following Figure 3.

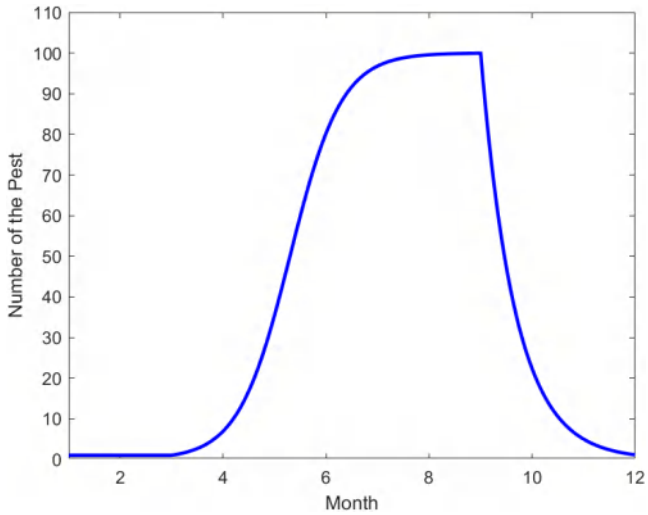

Fig. 3 The graph of hornets' number over time

As can be seen from Figure 3, the number of hornets increased rapidly from the beginning of the year and peaked around August. After that, due to climate disturbance, the number of hornets declined sharply until the end of the year. Thus, we can get the rules of hornet's quantity with the passage of time: the quantity increased first and then decreased, roughly showing a periodic change.

4 Hornet classification and priority investigation decision

4.1 Preparation of the model

This problem requires us to use only the data set file and the image files provided by Attachments to create, analyze, and discuss a model that predicts the likelihood of a mistaken classification. The model is used to discuss the classification of hornets, and to analyze how to prioritize the investigation of the reports most likely to be positive sightings. Obviously, this question is a problem of dichotomy. Therefore, we will establish a cost-sensitive RF model to classify the hornets in the data table (i.e., distinguish between Positive ID and Negative ID). First, before establishing the cost-sensitive RF model, we extract the characteristic indexes of hornets in the given 3305 images through image recognition, then combine the index information in the data table given by 2021MCM_ProblemC_DataSet.xlsx (which can be downloaded from the web page: http://www.mathismore.net/mcm/2021MCM_ProblemC_Files.rar, a password is required to open the file: Af6SP7rdm33PxPJmDb4wZq7cw) in Attachments to construct a complete index system of hornet classification. Second, we use the cost-sensitive method to convert the original unbalanced data set into a balanced data set. Finally, we establish an Asian giant hornet classification model based on the RF model. If the hornets in the reports were unable to identify, but they are identified as Asian hornets by our classification model proposed in this paper, then the corresponding reports needs to be priority investigation.

4.2 Index extraction

To classify the hornets, we must find out the indexes on which the classification is based. The question requires classification using only the data set file and the image files provided in Attachments, so we extract the classification indexes of the hornets from the given images and data tables.

4.2.1 Image recognition and feature extraction

Combining the rar file with 3305 images submitted with the sighting reports, we select the aspect ratio and color of the hornet's physical characteristics as two important characteristic indicators. Among them, the aspect ratio represents the ratio of the length to the width of the hornet. According to the data, the aspect ratio of the Asian giant hornet is generally more than 1.5 and its colors include the color of the head, chest, abdomen and tail tip. According to background of the problem, the Asian giant hornet has a yellow head, black chest, yellow and black or brown striped abdomen.

Aiming at the five indicators, i.e., aspect ratio, the color of head, the color of chest, the color of abdomen and the color of tail tip, we recognize the given images. If a hornet meets a certain index requirement, its index value is 1, otherwise it is 0.

4.2.2 Information extraction from data table

Combining the indicator information in the 2021MCM_ProblemC_DataSet.xlsx data table given in Attachments, we select the positive degree of eyewitness report tone and the location of the sighted hornets as the other two important indicators for hornet classification. Among them, the positive degree of eyewitness report tone indicates the certainty of the declarative sentence of the reporter, and the location of the sighted hornets is divided into longitude and latitude. According to the analysis result of the Problem 1, the main spread range of Asian giant hornet is 48.60° N to 49.20° N and 121.00° W to 124.80° W.

Similar to the previous method, if the location of a hornet is sighted within the range of 121.00° W to 124.80° W or 48.60° N to 49.20° N, the index value of longitude or latitude is assigned 1, otherwise it is 0.

For the judgment method of report tone, we use the LDA model in NLP natural language processing for machine learning (Chen et al. 2015). LDA is an artificial intelligence model that can process natural language. We input all declarative sentences into the LDA model to get the certainty grade of the tone of each sentence. The index values of the corresponding grades are specified in Table 1.

Table 1 The judgment method of the positive degree of eyewitness report tone

The positive degree of report tone	Grade	Index value
Very sure	A	1
General	B	0.5
Not sure	C	0

In summary, a total of 8 hornet classification indicators are involved. Thus, we can obtain the framework diagram of the indicator system of hornet classification, as shown in the following Figure 4.

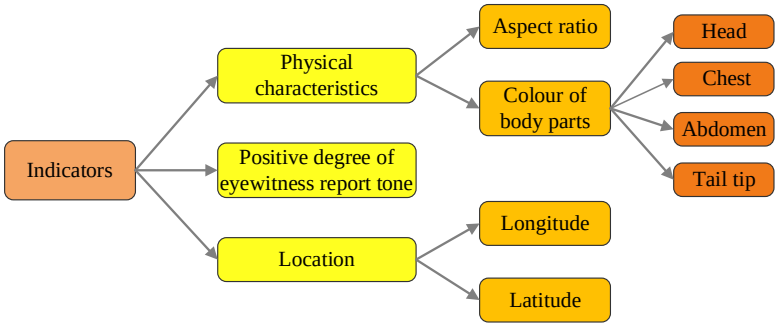

Fig. 4 The indicator system of hornet classification

Next, we establish a cost-sensitive RF model to classify the Asian giant hornet from all the reports in the 2021MCM_ProblemC_DataSet.xlsx data table given in Attachments.

4.3 The cost-sensitive RF model

Based on the statistics of 4440 report records in the 2021MCM_ProblemC _DataSet.xlsx data table given in Attachments, it is found that there are only 14 reports are Asian giant hornets, while there are 2069 reports confirmed to be other kinds of hornets and 2342 reports are unsure (A total of 4425). It can be obtained that the number of Positive ID: the number of Negative ID=14: 2069=1: 147. Obviously, the data set is seriously unbalanced. Therefore, we must first process the unbalanced data before establishing the classification model, that is, transform the unbalanced data set into a balanced data set. The conversion methods of unbalanced data set are divided into the data level and the algorithm level. In this paper, we introduce the cost-sensitive method from the algorithm level to transform the unbalanced data set into a balanced data set (Rao et al. 2020a; Yang et al. 2018; Yang et al. 2019b).

4.3.1 Cost-sensitive method

Cost-sensitive refers to the huge loss caused by misclassifying a certain category A into B in the classification problem. The solution is to construct a cost function (Yang et al. 2019b). The specific steps are as follows:

Step 1: Calculate the data center of each data category.

For the data set matrix

$$\begin{pmatrix} y_{11} & y_{12} & \cdots & y_{1m} & b \\ y_{21} & y_{22} & \cdots & y_{2m} & b \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ y_{n1} & y_{n2} & \cdots & y_{nm} & b \end{pmatrix}, \tag{25}$$

where each row is a hornet data sample, the data in each column are the indicator values, and the last column b is the classification of the data sample. In this paper, the data set has n (n=4425) data samples and m (m=8) data indicators. If this data set is only a majority data set or a minority data set, then the b in the last column is the same. If this data set is a majority data set b_1, then its center is

$$B_p = \frac{1}{n}\sum_{i=1}^{n} y_{ip}, \ \ p = 1, 2, ..., m, \tag{26}$$

i.e., the center of the majority data set b_1 is

$$(B_1, B_2, ..., B_m). \tag{27}$$

In the same way, the center of the minority data set b_0 and the entire data set N can be obtained.

Step 2: Calculate the weighted distance from the center of each category to the center of the entire data set.

We use the entropy method to calculate the weight of the data indicators in each category, i.e., the importance of the indicators in the data set.

For the indicator matrix

$$\begin{pmatrix} y_{11} & y_{12} & \cdots & y_{1m} \\ y_{21} & y_{22} & \cdots & y_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ y_{n1} & y_{n2} & \cdots & y_{nm} \end{pmatrix},$$

standardize the indicator values as follows:

$$Y_{ij} = \frac{y_{ij} - \min\limits_j y_{ij}}{\max\limits_j y_{ij} - \min\limits_j y_{ij}}, i = 1, 2, ..., n, \ j = 1, 2, ..., m. \tag{28}$$

For the negative indicators, the positive standardization is made as follows.

$$Y_{ij} = \frac{\max\limits_j y_{ij} - y_{ij}}{\max\limits_j y_{ij} - \min\limits_j y_{ij}}, i = 1, 2, ..., n, \ j = 1, 2, ..., m. \tag{29}$$

The entropy of each indicator is

$$F_j = \frac{\sum\limits_{i=1}^{n} p_{ij} \ln p_{ij}}{\ln n}, \tag{30}$$

where

$$p_{ij} = \frac{Y_{ij}}{\sum\limits_{i=1}^{n} Y_{ij}}. \tag{31}$$

Then the weight of indicators in each category is

$$w_j = \frac{1-F_j}{\sum_{j=1}^{m}(1-F_j)}, \tag{32}$$

which satisfies

$$\sum_{j=1}^{m} w_j = 1.$$

Let the weight of the majority data set b_1 be $w=(w_1,w_2,...,w_m)$, and the weight of the minority data set b_0 be $w^{'}=(w_1^{'},w_2^{'},...,w_m^{'})$, then the weight distance formula is

$$h_i = \sqrt{((B_j-\overline{B})\Omega)\Sigma^{-1}((B_j-\overline{B})\Omega)^{-1}}, \tag{33}$$

where $\Omega = diag(w_1,w_2,...,w_m)$, B_j is the center of the majority category, the corresponding $B_j^{'}$ is the center of the minority category, and $\overline{B}$ is the center of the entire data set.

Step 3: Define the coefficient of γ.

$$\gamma_i = \frac{\sum_{j=0}^{1} N_j}{N_i}, \tag{34}$$

where N_0 and N_1 are the sample numbers of minority category b_0 and majority category b_1 respectively. In this paper, it can be obtained that

$$\gamma_1 = 2045/2031,$$

and

$$\gamma_2 = 2045/14.$$

Step 4: Construct the cost function.

$$F(c_i,c_j) = \begin{cases} \gamma_i \dfrac{h_i^{'}}{h_j^{''}},\ h_i^{'} < h_j^{''} \\ \gamma_j \dfrac{h_i^{'}}{h_j^{''}},\ h_i^{'} > h_j^{''} \\ 0,\ i=j \\ 1,\ h_i^{'} = h_j^{''} \end{cases}, \tag{35}$$

where $h_i^{'}$ is the weight distance between category b_i and the entire data set *N*, and $h_j^{'}$ is the weight distance between category b_j and the entire data set *N*.

After calculation, the weights of the 8 indicators and the corresponding indicator data set centers are

shown in Table 2.

Table 2 The weights of indicators and the corresponding indicator data set centers

Indicator	Weight	Data set center
The color of head	0.0909	0.624
The color of chest	0.0909	0.675
The color of abdomen	0.0909	0.588
The color of tail tip	0.0909	0.534
Aspect ratio	0.1818	0.426
Positive degree of eyewitness report ton	0.2723	0.328
Longitude	0.0909	0.782
Latitude	0.0909	0.803

4.3.2 Cost-sensitive random forest

In this section, we establish a hornet classification model based on the random forest (RF) (Mercadier and Lardy 2019). We take the cost-sensitive decision tree as the base classifier, construct the cost function from the weight vector, split the nodes according to the indicator's cost reduction value REC as the standard, and use the accuracy rate weighted voting to judge the category. The specific steps are as follows:

Step 1: Construct the cost function and calculate the misclassification cost $F(b_1,b_0)$ and $F(b_0,b_1)$.

In this paper, $F(b_0,b_1):F(b_1,b_0)=14:2069=1:147.8$ and $F(b_i,b_i)=0(i=0,1)$.

Step 2: The original data set is sampled by *Bagging* to *k* training subsets are obtained.

Step 3: For each training subset:

(1) Randomly select *m* indicators from the original data set;

(2) Calculate *REC* in the feature subset, that is

$$REC = Nc - \sum_{i=0}^{n} Nc(B_i)\,, \tag{36}$$

where *Nc* is the cost value before splitting, *n* is the negative example contained in the current node, *FP* is the incorrect classification which is judged as correct, $\sum_{i=0}^{n} Nc(B_i)$ is the misclassification cost of choosing B_i splitting. Attribute *B* has *n* values, i.e., *B* has *n* different split nodes, and the sum of each learning cost is the cost of *B*.

(3) The largest REC node is selected each time to split to generate a decision tree without pruning.

Step 4: Record the classification accuracy rate of each decision tree is *q*, as the weight of the decision tree.

Step 5: For the test set and prediction set, weight the decision tree, and perform base classifier voting for prediction.

Finally, the output of the random forest is

$$G_c(x) = \arg\max_{y} \sum_{k} q_k I_{\{h_k(x)=y\}}\,, \tag{37}$$

where q_k is the weight of the decision tree, and *I* is the indicative function.

The flowchart of cost-sensitive random forest is shown in the following Figure 5.

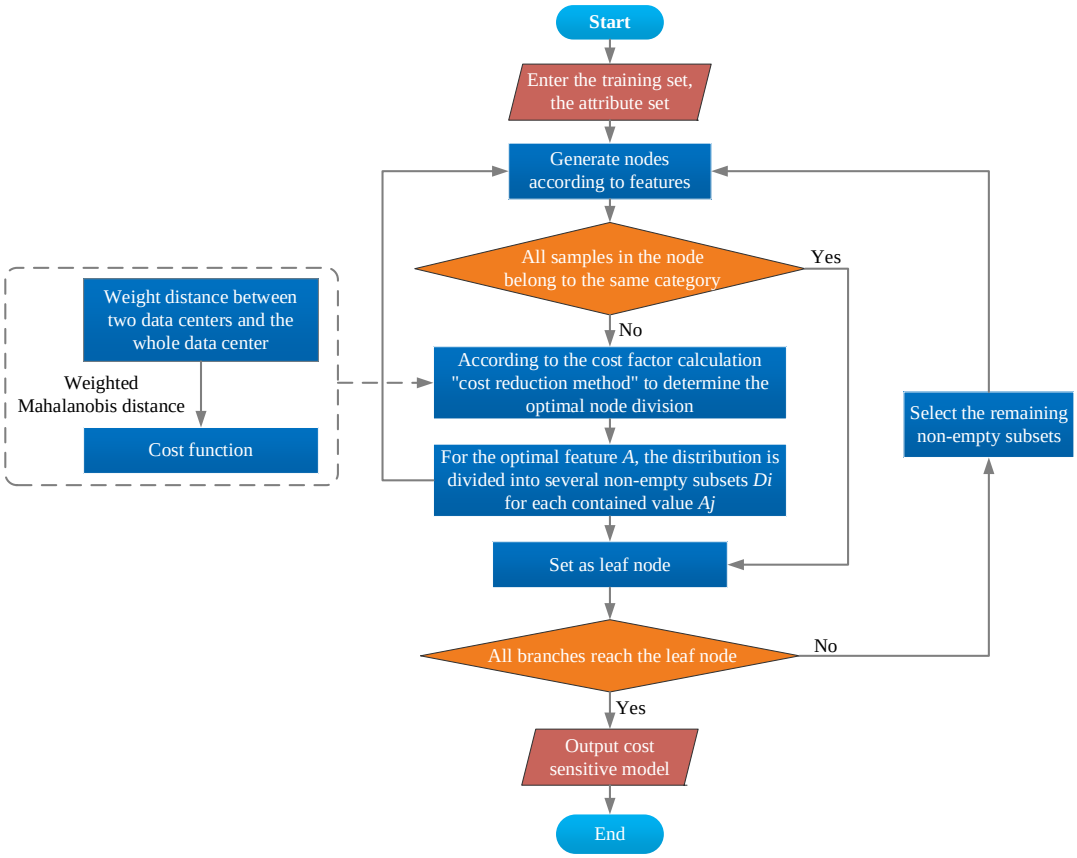

Fig. 5 The flowchart of cost-sensitive random forest

4.3.3 Solution of the model

We integrate 14 correct reports and 2069 error reports, and randomly select several reports to be substituted into the random forest for judgment. When the number of selected indicators is 3 to 8, the graph of the correct rate of the judgment results with increasing reports is as shown in Figure 6.

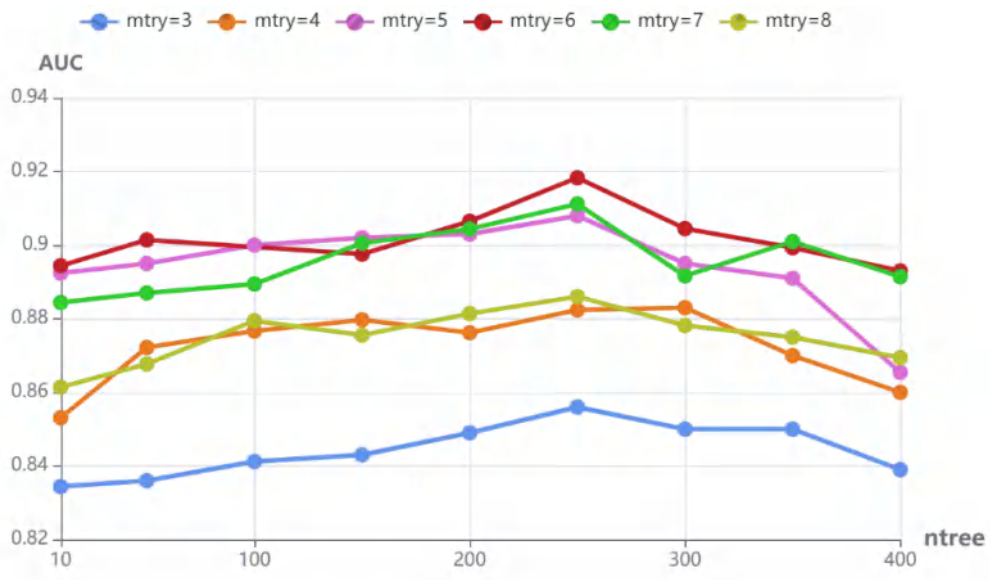

Fig. 6 The graph of the correct rate of judgment with increasing reports

4.4 Priority investigation decision

We substitute all the 2342 unprocessed reports into the cost-sensitive RF model established in Section5.3 for discrimination, and 324 correct reports (i.e., the Asian giant hornet) are identified. The results are shown in Table 3.

Table 3 New report records of the Asian giant hornets

Number	Global ID
1	{E6ADE6FB-0BD3-43EC-8E75-72EFC6F029FB}
2	{22E3A08D-494C-4539-8894-FDC32F2C9855}
3	{DA2999E6-B8F3-4BE9-B9E3-CB52F8F1C1DC}
4	{EF5051F6-1E6D-4A21-8F1C-A045E3DA56B4}
5	{57F0384B-53AE-4A4E-A417-951868E94C30}
…	…
323	{9561FAD1-C905-4890-B682-0478E859210D}
324	{2547522B-0531-48F3-A5BD-15A34E6C1E5A}

According to the conclusion obtained by Section 5.3.3, the accuracy of the judgment reaches 0.918, which is relatively reliable. In order to make the investigation more effective, the government should prioritize the investigation of the hornets mentioned in the 324 reports in Table 3, because the hornets in them are highly likely to be Asian giant hornets, that is, the reports are likely to be positive sightings.

5 Report update and Pest eradication certificate

5.1 Preparation of the model

This problem requires us to use the previous model to explain how to update when given additional new reports, as well as the frequency of the updates and evidence that the pest has been eradicated. Since we have studied the changes in the quantity of Asian giant hornet over time in Problem 1, we add human control factors and cycle parameters to further in-depth discussion on the basis of the Logistic model established in Problem 1.

5.2 Report update

5.2.1 Improved Logistic model

For this problem, we still take a nest as an example. Because it is difficult for hornets to survive in cold weather, we only conduct research in this problem from March to August. As the quantity of hornets gradually increases, the range of their activities will gradually expand, making them more visible and likely to be reported to the Department of Agriculture. Assuming that the number of times that people witness and report hornets is R, and its influence coefficient is σ=0.4. When a certain number is reached, people will report; but when it is not reached, there will be no impact on the hornets. The reporting threshold is set as N_θ, then the hornet's retardation factor becomes

$$(1-N/N_{\max}-(\sigma N/N_\theta)RI\{N\geq N_\theta\}), \tag{38}$$

where I is an indicative function:

$$I_{\{N\geq N_\theta\}}=\begin{cases}1, N\geq N_\theta\\ 0, N<N_\theta\end{cases}, \tag{39}$$

so the hornet population model is

$$\begin{cases} \dfrac{dN}{dt} = rN(1 - \dfrac{N}{N_{\max}} - \dfrac{\sigma N}{N_\theta} RI_{\{N \geq N_\theta\}}) \\ N(0) = 1 \end{cases}. \tag{40}$$

5.2.2 Solution of the model

The separation of variables method is used to solve the model (39), and the general solution is obtained as follows.

$$N(t) = \frac{N_{\max} N_\theta}{N_\theta + \sigma R N_{\max} I_{\{N \geq N_\theta\}} + (N_{\max} N_\theta - N_\theta - \sigma R N_{\max} I_{\{N \geq N_\theta\}}) e^{-rt}}. \tag{41}$$

Through continuous attempts, we found that when $R = 10$, $N_\theta = 30$, the results shown in Figure 7 below can be obtained.

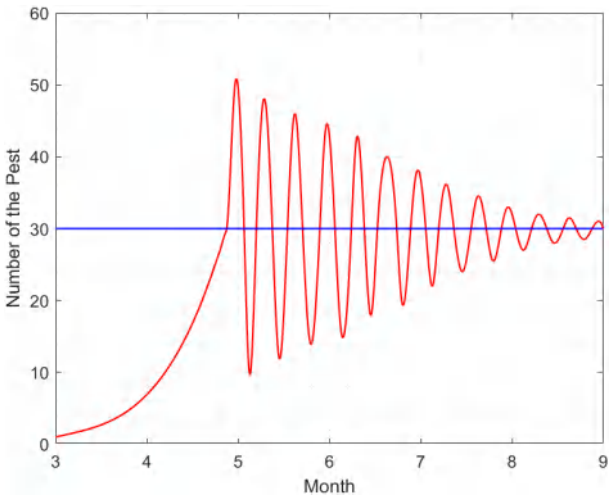

Fig. 7 The results of the improved Logistic model

According to Figure 7, we find that the population size of hornets fluctuates around the threshold over time, and there is a periodic phenomenon. We stipulate that over time, when the number of hornets exceeds a certain value, the additional reports will be updated, while the number is below this value, no updates will be made. According to Figure 7, it can be seen that the update report cycle is about once every 10 days.

5.3 Pest eradication certificate

Through the improved Logistic model established in Section 6.2 and its solution results, we find that when increase the human influence and lower the threshold, the number of Asian giant hornets will hardly increase over time. If the number fails to reach the threshold for a long time, that is, no report update is performed, or no hornet is found for a long time (such as 3 years), it means that the pest has been eradicated.

6 Model analysis and test

6.1 Goodness of fit test for fitting and grey prediction

The statistical interpretation of goodness of fit is the degree to which the regression function fits the observed value, and the coefficient of determination R^2 is a statistic to measure the degree of fit.

For a set of data *y* to be fitted, the mean is denoted as $\bar{y}$, and the fitting value is denoted as $\hat{y}_i$, then

$$SST = \sum_{i=1}^{n} (y_i - \overline{y})^2 \ , \tag{42}$$

$$SSR = \sum_{i=1}^{n} (\hat{y}_i - \overline{y})^2 \ , \tag{43}$$

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \ , \tag{44}$$

where n is the number of data, SST is the total sum of squares, SSR is the regression sum of squares, and SSE is the sum of squares of residual errors, then

$$SST = SSR + SSE \ . \tag{45}$$

The coefficient of determination is

$$R^2 = \frac{SSR}{SST} = \frac{\sum_{i=1}^{n} (\hat{y}_i - \overline{y})^2}{\sum_{i=1}^{n} (y_i - \overline{y})^2} = 1 - \frac{SSE}{SST} \ . \tag{46}$$

The maximum value of R^2 is 1, and the higher the determination coefficient is, the better the fitting degree is. Otherwise, the goodness of fit is worse.

By substituting the results of fitting model established in Section 4.2.1 and the improved grey prediction model established in Section 4.4.2 into the above test formulas (41)-(45), the test results can be obtained, as shown in Table 4.

Table 4 Test results of goodness of fit

	Fitting model	Grey prediction model
SST	1.7048	0.8040
SSR	1.6797	0.7603
SSE	0.0251	0.0437
R^2	0.9853	0.9456

It can be seen from Table 4 that the determination coefficients of the fitting model and the grey prediction model respectively reach 0.9853 and 0.9456, which means that both models have high goodness of fit.

6.2 Sensitivity analysis of Logistic model

In the Logistic model of studying the spread quantity of Asian giant hornets, we assume that the maximum hornet population of a honeycomb is $N_{\max} = 100$. In order to test the reliability of the spread rule, we observed the change of the Logistic function curve by changing the maximum number of the hornet population under the same other conditions. The results are shown in the following Figure 8.

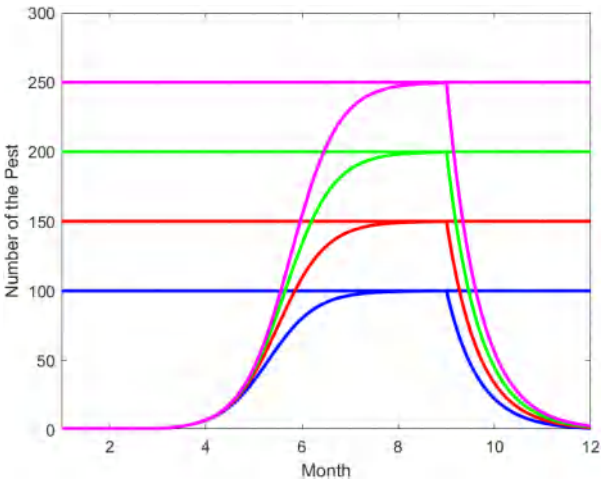

Fig. 8 Changes in hornet's population

From Figure 8, we can find that with the change of the maximum number of hornet population, the change rule of the number of hornet's population is firstly increased and then decreased. Therefore, through the sensitivity test, it can be known that the maximum value of the hornet population $N_{\max}$ will not disturb and affect the trend of the final predicted results.

6.3 Test for the Cost-sensitive RF model

In this paper, when establishing the cost-sensitive RF model, we integrate the cost-sensitive method to carry out the conversion processing of unbalanced data. In order to test the effect of this model, we compared the results of our cost-sensitive RF model with that of the traditional RF model, and the comparative indexes of model performance evaluation are as follows:

(1) Classification accuracy *Ac*: the proportion of correctly classified samples to the total number of samples, which is expressed as

$$Ac = \frac{TP + TN}{TP + TN + FP + FN}, \tag{47}$$

where *TP* represents the number of samples originally judged to be correct in the actual classification, *TN* represents the number of samples originally judged to be wrong in the actual classification, *FP* represents the number of samples originally judged to be correct in the actual classification, and *FN* represents the number of samples originally judged to be wrong in the actual classification.

(2) Accurate rate *p*: the proportion of correctly classified samples in the positive sample after prediction. The formula is

$$P = \frac{TP}{TP + FP}. \tag{48}$$

(3) Recall rate *R*: the proportion of correctly classified samples in the positive category, and the formula is

$$R = \frac{TP}{TP + FN}. \tag{49}$$

(4) *Kappa* coefficient: an index to judge the accuracy of classification and to judge whether different methods are consistent in predicting results. The formula is

$$Kappa = \frac{P_0 - P_e}{1 - P_e}, \tag{50}$$

where

$$\begin{cases} P_0 = \dfrac{TP+TN}{N} \\ P_e = \dfrac{A\times C+B\times D}{N^2} \end{cases}. \tag{51}$$

(5) *G-mean*: the performance of evaluating unbalanced data, and the formula is

$$G-mean=\sqrt{\frac{TP}{TP+FN}\times\frac{TN}{TN+FP}}. \tag{52}$$

(6) *F1-measure*: which is used to evaluate the classification of unbalanced data. The higher its value is, the better the classification effect is. The formula is

$$F1-measure=\frac{2\times P\times R}{P+R}. \tag{53}$$

After calculation, the comparison results between the cost-sensitive RF established in this paper and the traditional RF are shown in the following Table 5.

Table 5 Performance evaluation results of the two types of RF models

	Cost-sensitive RF	Traditional RF
Ac	0.9306	0.9222
P	0.0676	0.0440
R	0.7143	0.5000
Kappa	0.1124	0.0692
G-mean	0.8159	0.6801
F1-measure	0.1235	0.0809

It can be clearly seen from the performance evaluation results in Table 5 above that the six indexes of cost-sensitive RF are better than the traditional RF. Therefore, the introduction of cost-sensitive method to process the unbalanced data plays a good effect of improving the accuracy of classification results.

7 Conclusions

In order to correctly identify Asian giant hornets and effectively inhibit its wantonly spread, the improved grey prediction model, the Logistic model and the cost-sensitive random forest model were established in this paper to systematically study the spread and classification of Asian giant hornets.

The contribution of this paper lies in: (i) This paper studied the propagation law of hornets from the perspective of the propagation range and the number of hornets. On the one hand, by fitting the correct reported latitude and longitude, and establishing an improved grey prediction model, we systematically predicted the latitude and longitude of hornets with changes of time. On the other hand, by establishing the improved Logistic model and combining with the function image, we got the number of hornets with changes of time. (ii) In order to solve the dichotomy problem of the hornet based on texts and images, the traditional cost-sensitive random forest model is improved and a new fusion cost-sensitive random forest model is presented to deal with the problem of unclassified bumblebee binary classification. This model is better solved the poor classification results of the problems of unbalanced data sets. Moreover, it improves the adaptability and robustness of the original classifier, and has a better classification effect for the classification of unbalanced data sets. (iii) To report update and pest eradication and other issues, we added artificial control factors and cycle parameters based on the law of diffusion, and established an improved Logistic model. The result shows that the population size of the hornet time fluctuation near the threshold,

and the cyclical phenomenon exist. Thus, we gave the updating cycle of reported and the conditions of judgment pest eradication.

The large data sets generated by the Internet are often high-dimensional and noisy. Thus, the operation of ensemble learning algorithm usually takes a long time, and this kind of algorithm needs further research and optimization. In the future work, we will fully consider the influence of high-dimensional and multi-noise data sets, and establish some new models based on the model proposed in this paper, which can be used for the classification of high-dimensional and multi-noise unbalanced data.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants performed by any of the authors.

Informed consent Informed consent was obtained from all individual participants included in the study.

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大学生互联网理财参与情况及影响因素探究
——以武汉市为例

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摘 要：本研究通过对武汉市 8 所高校大学生进行问卷调查，建立 logistics 回归和决策树模型对 1096 份有效问卷进行数据分析，得到以下关于大学生互联网理财参与情况及影响因素的研究结论：大学生群体的互联网理财参与情况倾向积极但仍有待提高；学历、专业、月可支配金额、感知价值和社交圈的理财情况对大学生互联网理财参与情况有正向影响，感知风险对理财参与情况有负向影响，其中月可支配金额的影响程度最大，其次为感知价值、专业、感知风险、学历和社交圈的理财情况。据此，本文针对学校、家庭、大学生投资者和互联网理财平台这四个不同主体分别提出合理化建议，以期共同促进大学生更积极、更安全地进行互联网投资理财。

关键词：大学生；互联网理财；影响因素；logistics 回归；决策树

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Research on College Students' participation in Internet financial management and its influencing factors

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Abstract: Based on the questionnaire survey of college students from 8 universities in Wuhan, this paper establishes the logistic regression and decision tree model to analyze the data of 1096 valid questionnaires, and obtains the following research conclusions about college students' participation in Internet financial management and its influencing factors: College Students' participation in Internet financial management tends to be positive, but it still needs to be improved; The influencing factors of College Students' participation in Internet financial management are education background, major, monthly disposable amount, desire for income, worry about risk and financial situation of the people around them. The influence of monthly disposable amount is the largest, followed by desire for income, major, worry about risk, education background and financial situation of the people around them. Finally, based on the above conclusions, this paper puts forward reasonable suggestions for four different subjects, namely, school, family, college students' investors and Internet financial platform, in order to jointly promote college students' more active and safer Internet investment and financial management.

Keywords:college student;Internet finance;influencing factors;Logistic regression; Decision tree

一、引言

受新冠疫情影响，全球经济在经历了 2020 年年初的震荡、被压制后逐渐实现稳步回升，领涨全球，带动了 3 月投资者数量的激增。6、7 月 A 股市场行情火爆迎来快速上涨，上证指数突破 3000 大关，再次点燃投资者入市热情。从 2020 年底“基金”话题频上热搜，平台、大 V、主播等多方合力联动，到 2021 年前各板块一路领涨再创历史新高，将“后疫情时代的基金热”再度推向一个高潮。在这样的形势下，越来越多的年轻人紧跟潮流，下海“养基”，希望凭借自己的力量“发家致富”。与此同时，2020 年 8 月上海高金金融研究院发布报告显示，中国理财群体呈现明显的年轻化，年轻人正在逐渐成为理财的“主力军”（2020 国人理财趋势报告）。大学生是年轻人中的重要特殊群体，思想观念深受互联网影响的同时也正在直接或间接地接受理财有关的教育、未来将进入社会各行各业中工作，这为互联网理财的进一步扩大化、全民化提供了充沛的、可发展的“潜力股”与“后备军”。那么，当今大学生互联网理财的参与情况如何？又是什么因素在影响大学生参与互联网理财的热情？针对上述问题的深入研

究，能发现大学生在互联网理财方面上存在的规律和问题，从而给个人、家庭、学校及平台提供可行性建议，对互联网理财的宏观发展与所涉主体的微观调控有着重要的理论与实践意义。

在已有的研究中，一些学者从基本情况（性别、年龄、专业、月收入等）出发，结合了问卷调查法，对影响人们互联网理财参与情况的因素进行了实证分析[1]；还有一些学者从社群互动视角、心理因素视角、金融素养视角、初始信任视角等方面分析了人们理财决策的影响因素。然而，目前针对年轻群体，尤其是高校大学生的深入研究相对较少，并且考虑的影响因素较为片面，运用的研究方法比较单一。本文立足高校大学生视角，将影响因素分为个体背景、效价理论和社会影响理论共三个层面，以武汉市大学生为对象进行实证研究，在传统的文献研究、问卷调查、深度访谈等研究方法的基础之上，综合运用 logistics 回归和决策树模型对调查数据进行数据分析，丰富了国内现有的关于互联网理财参与情况及其影响因素的研究。

二、理论分析与研究假设

1. 互联网理财的基本内涵与理论研究

现阶段互联网理财没有一个公认的、成熟的概念界定。2016 年，赵保国等将网络投资理财定义为“个人审视自己的资产状况及风险承受能力，利用网络投资工具合理运用财务资源，实现财产的保值增值”。该定义中的网络投资理财工具指基于互联网特色建立的、仅能通过互联网渠道购买的在线理财产品。另外，中国互联网络信息中心定义互联网投资理财产品为：“互联网公司或银行发行的，只能通过互联网渠道购买的余额宝类理财产品，该类产品具有高收益、低门槛、高流动性等特点”。

互联网理财产品本质仍为理财产品，但同时具有门槛低、灵活性高、收益高、操作方便快捷等特点，所以用户的互联网理财参与情况在考量理财行为本身属性的同时，受到情绪变动、社会环境影响的程度相较于传统理财行为更加明显。因此，本文选择结合基于主观因素影响的效价理论与基于外部影响的社会影响理论，对大学生互联网理财参与情况的影响因素进行分析。与此同时，针对大学生这一特定群体进行研究与挖掘时，群体内部的个体基本差异、具有群体特色的教育背景差异等群体内个体背景差异对其互联网理财参与情况也是值得细化考量的重要影响因素。综上，本文依据效价理论与社会影响理论两个维度，结合群体中个体背景的主要差异，对大学生互联网理财参与情况进行分析，以期通过实证研究探索出大学生互联网理财参与决策的影响因素。

2. 个体背景与理财参与情况

对大学生这一群体进行互联网理财行为的深入研究，需要将群体内部的个体基本差异（性别、户籍等）与具有群体特征的教育背景差异等纳入重点考虑范畴，作为自变量而非控制或调节变量，以凸显群体特征、对群体进行更加细致且深入的刻画。Robb 和 Sharpe 指出，学历与金融知识对于高校学生的理财行为有着重要的影响。Veld—Merkoulova 则研究了投资者背景对于个人投资决策的影响情况。大学生群体内部个体的背景，主要涉及基本差异与教育背景差异。基本差异主要包括性别、户籍所在地（城乡）、月可支配金额等。教育背景差异反映了大学生特殊的群体特征，主要包括就读中的学校层次、学历、专业等。基于此，本文提出如下假设：

H1：性别会对大学生互联网理财参与情况产生影响；

H2：户籍所在地会对大学生互联网理财参与情况产生正向影响；

H3：月可支配金额会对大学生互联网理财参与情况产生正向影响；

H4：学校层次会对大学生互联网理财参与情况产生正向影响；

H5：学历会对大学生互联网理财参与情况产生正向影响；

H6：专业会对大学生互联网理财参与情况产生正向影响。

3. 效价理论与理财参与情况

1975 年，Peter 和 Tarpey 提出了效价理论，认为人们在进行决策时的感知有正、负两种属性，即感知价值与感知风险，分别对人们的决策有正效用与负效用。感知正效用与感知负效用是人们进行决策的行为基础，是效价理论的核心所在。本文认为，感知价值与感知风险会明显影响大学生互联网理财的参与情况。

① 感知价值

感知价值指用户做出决策前所感知到该项决策做出后可获得的价值。Zeithaml 认为，感知价值是用户基于获得与付出的感知，是对效用的综合性评价。Baker 等人认为，感知产品价值是消费者在进行购物活动时产生的感知利得和感知成本间的权衡，与未来购买意愿呈高度相关。Petrick 通过实证研究验证了服务业中顾客感知价值与购买行为之间的正相关关系。李克会以网络团购为例进行研究发现，消费者情感反应在网站特征、感知价值与冲动性购买意愿之间起中介作用。感知价值在用户进行互联网理财参与决策时具有非常重要的影响。基于此，本文提出如下假设：

H7：感知价值会对大学生互联网理财参与情况产生正向影响。

② 感知风险

Bauer 在 1960 年将感知风险引入到消费者研究中，此后很多学者将感知风险视为用户行为研究中的一项重要因素。2016 年，赵保国将感知风险引入互联网理财领域，并将网络投资理财语境中的感知风险定义为“个人用户在网络投资理财中，对预计可能发生的、与开始期望不相符的负面结果的心理预期”，同时将风险关注重点集中在非系统性风险。

个人用户参与互联网理财的目的是实现资产增值，然而风险不可避免。个人用户在参与互联网理财前会对存在的风险进行评价。一旦评价者认为可能会出现损害自身利益的风险，那么就会产生抵触情绪而负向影响参与决策。用户在参与互联网理财之前便会特别注重如何降低理财过程中的风险，其自身衡量的感知风险在用户进行互联网理财参与决策时非常重要。基于此，本文做出如下假设：

H8：感知风险会对大学生互联网理财参与情况产生负向影响。

4. 社会影响理论与理财参与情况

社会影响理论常用于解释个人或者组织对于个体行

为产生的影响。社会影响指个体在他人或群体的影响下思想、情感与行为的发生和变化，反映出个体的态度、信念和行为受外界影响的程度。Kelman 多次研究不同情境下个人因受到他人或组织的影响而发生的态度与行为的变化；李振鹏与唐锡晋基于社会影响理论分析了群体整体观点与系统外因素共同影响个体间的互动使其演化为不同的模式。基于互联网的低门槛及灵活性和社交圈的外在干预，本文将基于社会影响理论，来探讨其对大学生互联网理财参与决策的影响作用。基于此，本文提出如下假设：

H9：社交圈的理财情况会对大学生互联网理财参与情况产生正向影响。

三、研究设计

1. 样本来源

本研究在大量文献阅读和相关理论学习的基础上，采用网络爬虫技术和线下深度访谈调研，对各大平台投资理财相关搜索进行词频分析，并对大学生互联网理财行为进行初步了解，设计了预调查问卷。通过小组座谈会的方式对预调查问卷进行加工和修正，得到正式调查的问卷。调查问卷共分为四部分内容：第一部分为基本情况，涉及被调查对象的性别、户籍所在地、月可支配金额、学校层次、学历和专业；第二部分为互联网理财参与情况；第三部分为主观感受，涉及感知价值和感知风险；第四部分为外在干预，涉及社交圈的理财情况。

正式调查在 4 月开展，基于配额的分层二阶抽样，在武汉市共抽取 8 所高校，发放 1200 份问卷，回收有效问卷 1096 份，有效回收率达 91.3%，有效问卷的样本特征如表 1 所示：

表 1　样本特征

样本特征		所占比例
性别	男	50.1%
	女	49.9%
户籍所在地	城市	54.7%
	农村	45.3%
学校层次	985/211 院校	26.8%
	普通本科院校	45.0%
	专职类院校	28.2%
学历	专科	35.7%
	本科	46.2%
	硕士	14.0%
	博士	4.1%
专业	经管类	23.0%
	非经管类	77.0%
月可支配金额	1000 元以下	30.8%
	1000~1999 元	49.3%
	2000~2999 元	10.7%
	3000 元及以上	9.2%

2. 变量说明

根据之前的假设，本文设置大学生互联网理财参与情况为因变量，性别、户籍所在地、学校层次、学历、专业、月可支配金额、感知价值、感知风险和社交圈的理财情况，具体的变量符号及变量解释如表 2 所示。

表 2　变量说明

变量名称	变量符号	变量解释
理财参与情况	y	$y=\begin{cases}1,\text{愿意}\\0,\text{不愿意}\end{cases}$
性别	x_1	$x_1=\begin{cases}1,\text{男生}\\2,\text{女生}\end{cases}$
户籍所在地	x_2	$x_2=\begin{cases}1,\text{乡村}\\2,\text{城市}\end{cases}$
学校层次	x_3	$x_3=\begin{cases}1,\text{专职类}\\2,\text{普通本科}\\3,985/211\end{cases}$
学历	x_4	$x_4=\begin{cases}1,\text{专科}\\2,\text{本科}\\3,\text{硕士}\\4,\text{博士}\end{cases}$
专业	x_5	$x_5=\begin{cases}1,\text{非经管类}\\2,\text{经管类}\end{cases}$
月可支配金额	x_6	$x_6=\begin{cases}1,1000\text{元以下}\\2,1000—1999\text{元}\\3,2000—2999\text{元}\\4,3000\text{元及以上}\end{cases}$
感知价值	x_7	$x_7=\begin{cases}1,\text{无价值}\\2,\text{一般}\\3,\text{很有价值}\end{cases}$
感知风险	x_8	$x_8=\begin{cases}1,\text{无风险}\\2,\text{一般}\\3,\text{很有风险}\end{cases}$
社交圈的理财情况	x_9	$x_9=\begin{cases}1,\text{从未参与}\\2,\text{偶尔参与}\\3,\text{经常参与}\end{cases}$

3. 信度和效度检验

信度检验包括内部一致性信度（Cronbach's Alpha 系数）检验和组合信度(CR 值)检验。[6]经检验，各变量的内部一致性系数在 0.834~0.932 之间，均大于 0.8，且组合信度的 CR 值在 0.734~0.898 之间，均大于 0.7，说明变量具有良好的一致性和可靠性，问卷的信度良好。

效度检验包括内容效度检验和结构效度检验。通过 SPSS22.0 软件，得到各变量之间的相关系数 P 值在 0.752~0.826 之间，均在 0.7 以上，说明问卷的内容效度良好。其次，得到各变量的 KMO 系数为 0.913（大于 0.7），同时 Bartlett 球状检验的显著性水平小于 0.05，说明各个变量之间存在共享公因子的可能性低，故能进行因子分析。采用主成分分析法提取公因子，结果显示标准化的因子载荷值均高于 0.6，表明结构效度良好。

四、实证结果分析

1. 描述性统计分析

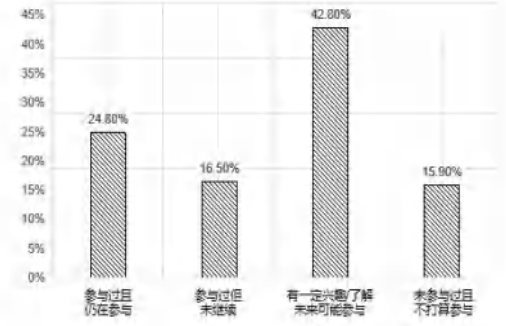

图 2　大学生互联网理财参与情况

由图 2 可知，在本次调查回收的 1096 份有效问卷中，参与过互联网投资理财的大学生占 41.3%，未参与过甚至不打算参与的大学生占 58.7%。可见，大部分大学生具备了一定的理财意识，但仍有部分同学因为各种原因不愿意参与其中。因此，大学生的理财意识仍有待提高。

2.logistics 回归分析

为了研究大学生互联网理财参与情况的影响因素，本文先建立 logistics 回归模型[3]，具体变量设置见表 1。接下来，通过逐步回归的方法筛选出对大学生互联网理财参与情况具有显著性影响的因素。

逐步回归法是一种选择最优量子集的方法，可以从提供的可供挑选的变量中，根据一定的显著性检验，每一步只选入一个变量进入方程，确保回归方程中只含有对被解释变量显著影响的解释变量。在本文中，为了得出真正影响投资意愿的变量，以显著性检验的 *P* 值作为信息准则筛选自变量，当变量显著性检验的 *P* 值<0.05 时，将该变量选入模型；否则，剔除该变量。[7]

结合 R 软件，得到逐步回归的结果如表 3 所示：

表 3　逐步回归结果

影响因素	回归系数	标准误	Wald χ^2	*P* 值	*OR* 值	95%CI
学历	0.331	0.159	7.361	0.007	0.650	0.476-0.887
专业	0.405	0.219	3.424	0.004	0.667	0.434-1.024
月可支配金额	0.442	0.113	13.852	0.000	1.525	1.221-1.904
感知价值	0.434	0.173	0.600	0.009	0.875	0.623-1.228
感知风险	-0.378	0.204	4.638	0.003	5.249	0.369-1.144
社交圈的理财情况	0.288	0.196	10.462	0.000	8.746	2.364-8.432
截距	0.688	0.712	14.262	0.000	14.701	3.643-59.320

由表 3 可以看出，对大学生互联网理财参与情况有显著性影响的因素有学历、专业和月可支配金额、感知价值、感知风险和社交圈的理财情况，而性别、户籍所在地和学校层次对大学生互联网理财的参与情况没有显著性影响。其中，学历、专业、月可支配金额、感知价值和社交圈的理财情况对大学生理财参与情况有正向影响，感知风险对理财参与情况有负向影响。因此，假设 H4、H5、H6、H7、H8 和 H9 通过验证，假设 H1、H2 和 H3 未通过验证。另外，根据表 3 中回归系数可以得出，对理财参与情况影响最大的因素为月可支配金额，其次是感知价值、专业、感知风险、学历及社交圈的理财情况。

3. 3. 决策树检验

为了进一步验证 logistics 模型中影响变量对理财参与情况的影响程度的排序，我们使用决策树模型来对影响变量进行重要性水平排序。

决策树中不纯性的度量使用基尼系数来计算，公式如下所示：

$$Gini(t) = 1 - \sum_{i=0}^{c-1} [p(i \mid t)]^2$$

其中 *c* 是类的个数，*t* 为所在的结点位置，*p(i|t)*为此结点中第 *i* 类的分布所占频率，基尼系数的取值范围为[0,1]。

影响变量的重要性的判断，我们用信息增益来表示，公式如下所示：

$$\Delta_{\inf o} = Gini(parent) - \sum_{j=1}^{k} \frac{N(v_j)}{N} Gini(v_j)$$

其中 *parent* 是父节点，*N* 是父节点上的记录总数，*k* 是属性值的个数，$N(v_j)$是与子女结点 v_j 相关联的记录个数。[8]

在 logistics 回归模型中，影响变量为学历、专业和月可支配金额、社交圈的理财情况、感知价值和感知风险，因变量为理财参与情况。因此，本文以理财参与情况作为根结点构建决策树，结合 SPSS 软件，得到影响变量的信息增益（即重要程度），结果见表 4。

表 4　决策树模型结果

	信息增益	排序
月可支配金额	0.174	1
感知价值	0.166	2
专业	0.158	3
感知风险	0.146	4
学历	0.126	5
社交圈的理财情况	0.114	6

由表 17 可知，影响变量重要性的排序为：月可支配金额、感知价值、专业、感知风险、学历和社交圈的理财情况，与 logistics 模型得到的影响程度排序一致，说明了模型变量筛选的正确性。

五、结论与检验

1.结论

①大学生群体的互联网理财参与情况倾向积极但仍有待提高

随着生活日渐充裕，大部分大学生具有一定的投资理财意识，近半数的大学生已经接触了互联网投资理财，而且大部分同学都对理财持积极态度，但仍有一部分同学不愿意参与互联网理财投资，理财意识仍有待提高。

②大学生互联网理财参与情况的影响因素

学历、专业、月可支配金额、感知价值和社交圈的理财情况对大学生理财参与情况有正向影响，感知风险对理财参与情况有负向影响。其中影响程度最大的因素是月可支配金额，其次是感知价值、专业、感知风险、学历及社交圈的理财情况。

2.建议

本次调查发现，要想更好的促进大学生参与到互联网投资理财当中去，需要各方形成合力，共同改善现存问题和不足。因此，本文针对学校、家庭、大学生投资者和互联网理财平台这四个不同主体，提出如下建议。

①学校应增强对大学生投资理财的宣传教育

研究表明，经管类专业的大学生的理财参与热情更高，为了调动更多大学生的互联网理财投资热情，学校应该开设并更好地建设面向全校的投资理财主题公共课。大学生群体不同于工作人士，需要进行专门的适合于学生群体的投资理财教育。另外，邀请校内或社会专业人士进行投资理财主题讲座，对学生进行课外知识的填补与拓展，使学生在投资理财方面得到更前沿、更实战的经验与知识，拓宽学生的知识范围与眼界、拔高投资理财能力。

②家庭应给予相应的支持并引导正确的理财观念

研究表明，大学生互联网理财参与热情会受社交圈理财情况的影响，而家庭显然是一个重要的主体。家庭是对一个人影响最大的一个环境，对孩子进行理财方面的教育、培养，是有着积极而深远的意义的。家庭需要引导正确的家庭投资理财观念，当一个家庭已经具备了正确的理财观念与完备的理财规划，理财的思想会自然流入孩子的人生中。家庭给予的正确、科学、恰当的理财教育，会使孩子从小建立起科学合理的理财意识。

③大学生投资者应当增强自己的理财意识

调查发现，大学生群体的互联网理财意识还有待加强。因此，大学生应当有意识地在投资理财实践中进行规律发掘与实操历练，增强理财素养。另外，积极进行相关专业知识的学习。例如，学习专有名词、理财规律、行业规律、技术手段等。将理论与实践相结合，进一步加深对互联网理财的认知。

④平台应加强安全建设，保护客户合法权益

经研究发现，风险是阻碍大学生参与投资理财的重要的因素，其中就包括了互联网理财平台带来的风险。因此，平台方作为人们进行投资理财运作的交通枢纽，应采取公平公正的态度，客观地呈现信息，合法合理地运营，且做到对用户信息与财产安全的严密保护。

第一，平台方在保护用户的财产安全的同时，也应尊重并维护用户的信息安全、严密保护用户的财产信息与安全。作为互联投资网理财的交易平台与大量用户重要信息的掌握者，平台方更应该做好自身的安全建设，承担保护用户的财产与信息安全的义务与责任。

第二，平台方应客观地进行理财产品信息的呈现，不对信息进行歪曲，不为了不当牟利与其他主体进行非法合作。作为交易时双方信息呈现的直接与最重要的平台，平台方应做到一个呈现信息真实、客观，同时在其他方面积极为用户提供理财中不同方面的服务第三方。做好自身的职业道德的建设，严格遵守行业的相关法律法规，坚决维护客户的合法权益。

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